Hrant Patrick Hratchian

Graduate Division
University of California, Merced
Merced, CA 95348

Office: 318 SSB
Email: hhratchian@ucmerced.edu

Administrative Experience

Vice Provost and Dean for Graduate Education — 2022–Present

Provides university-wide leadership for graduate education and postdoctoral affairs at the University of California, Merced, including graduate admissions and enrollment management, doctoral education strategy, graduate program development and assessment, academic policy development, graduate student and postdoctoral success initiatives, and Graduate Division budget planning and resource allocation. Collaborates across university divisions – including Academic Affairs; Student Affairs; Finance and Administration; Research & Economic Development; Enrollment Management; External Relations & Strategic Communications; and Equity, Justice, & Inclusive Excellence – as well as academic departments and graduate programs to advance institutional priorities related to student success, academic planning, research growth, and institutional transformation. Serves on the Chancellor's Extended Cabinet and Deans' Council and leads a division responsible for campus-wide graduate education operations and services. Serves ex officio on the Academic Senate's Graduate Council and represents UC Merced on the University of California Council of Graduate Deans and systemwide policy and advisory committees, including the UC Systemwide Committee on Residency Policy and the Systemwide Library and Scholarly Information Advisory Committee.

Selected Accomplishments

Institutional Strategy and Graduate Education

- Served as a key academic leader in UC Merced's successful advancement to Carnegie R1 classification, including sustained growth in doctoral education and research capacity. During tenure as Vice Provost and Dean, annual doctoral degree conferrals surpassed the Carnegie R1 threshold by more than 75%.
- Led implementation of a campus-wide doctoral enrollment planning process in collaboration with academic leadership, improving strategic alignment between graduate admissions, program capacity, funding models, and long-term institutional goals.
- Expanded the scale and impact of the Competitive Edge Summer Bridge Program, increasing the number of funded graduate student fellows by more than 100% over two years.

ADMINISTRATIVE EXPERIENCE

Vice Provost and Dean for Graduate Education

Budget and Organizational Leadership

- Maintained long-term fiscal sustainability while leading Graduate Division budget planning, strategic resource allocation, and graduate fellowship and award planning.
- Collaborated with the Provost's Office and Chief Financial Officer to redesign graduate program funding models through implementation of block grant allocation structures and increased institutional investment in graduate education.
- Led organizational restructuring of the Graduate Division, including creation of new leadership and staff positions, expanded operational capacity, and enhancement of student and program support services without reliance on new central campus funding.

Student Success, Research, and Partnerships

- Established UC Merced as an ARCS Foundation campus and expanded philanthropic support for graduate education, including approval of an additional participating department and doubling of ARCS support.
- Partnered with Academic Personnel and campus leadership to improve postdoctoral scholar recruitment, hiring, and support processes in alignment with the university's research growth strategy.
- Helped foster organizational stability and a positive graduate student, staff, and faculty climate during a graduate student union strike through transparent communication, collaborative problem-solving, and sustained engagement with campus stakeholders.

Chair, Department of Chemistry & Biochemistry 2021–2022

Provided academic and administrative leadership for the Department of Chemistry & Biochemistry, including strategic planning, budget and resource management, faculty recruitment and development, merit and promotion review processes, curricular oversight, student success initiatives, and degree program assessment within a rapidly growing R2 research university environment. Worked collaboratively with faculty and staff through shared governance to support the department's teaching, research, and service missions while advancing research productivity, educational quality, organizational effectiveness, and institutional priorities. Coordinated departmental planning, communication, and operational decision-making during periods of significant instructional and organizational disruption associated with the COVID-19 pandemic and broader campus labor and operational challenges.

ADMINISTRATIVE EXPERIENCE

Chair, Department of Chemistry & Biochemistry

Selected Accomplishments

- Led departmental budget planning and resource management in support of instructional, research, and operational priorities.
- Established new departmental governance and communication structures to support shared decision-making, faculty engagement, and organizational effectiveness.
- Developed and implemented community-building initiatives designed to strengthen student belonging and departmental climate.
- Coordinated instructional continuity and academic operations during the COVID-19 Delta and Omicron waves through development and implementation of temporary alternative educational and instructional plans.
- Developed departmental academic continuity and resilience plans in preparation for potential campus labor disruptions, working collaboratively with faculty, staff, and campus leadership to support instructional continuity and student success.

Interim Co-Director, Cyberinfrastructure & Research Technologies 2019–2021

Provided strategic and operational leadership for campus-wide computational research infrastructure and research technology services within the Office of Information Technology, including high-performance computing systems, advanced research networking, and visualization infrastructure supporting more than 600 researchers across UC Merced and partner institutions. Reported directly to the Chief Information Officer and served on the Office of Information Technology leadership team, working collaboratively with campus leadership, faculty, and research support units to advance research computing capacity, faculty research support, infrastructure planning, and long-term operational sustainability in support of the university's research mission. Provided leadership for professional staff and coordinated research technology operations during a period of significant institutional growth and operational disruption associated with the COVID-19 pandemic.

Selected Accomplishments

- Supported computational research infrastructure and services serving more than 600 researchers across UC Merced and partner institutions.
- Served as PI on NSF-funded infrastructure projects supporting deployment of the MERCED and Pinnacles campus-wide high-performance computing clusters, significantly expanding institutional research computing capacity.
- Collaborated with the Provost and Vice Chancellor for Research to develop sustainable operational and budget models for campus research computing infrastructure and services.

Administrative Experience

Interim Co-Director, Cyberinfrastructure & Research Technologies

Selected Accomplishments

- Led development of a Recharge Center component within CIRT to support long-term operational sustainability and scalable research support services.
- Developed processes and procedures supporting faculty recruitment and startup planning through early engagement between CIRT and faculty hiring initiatives.
- Led operational coordination and deployment of the Pinnacles high-performance computing cluster during COVID-19 emergency operational restrictions, ensuring continuity of critical research infrastructure expansion.
- Recruited and led professional staff supporting research computing operations and services, including leadership transition planning for the permanent Director appointment.

Appointments & Employment

2022 – Present	Vice Provost and Dean for Graduate Education University of California, Merced
2013 – Present	Professor (2023), Associate Professor (2020), Assistant Professor Department of Chemistry & Biochemistry, UC Merced
2021 – 2022	Chair Department of Chemistry & Biochemistry University of California, Merced
2019 – 2021	Interim Co-Director Cyberinfrastructure and Research Technologies Office of Information Technology University of California, Merced
2008 – 2013	Research Scientist Gaussian, Inc. Wallingford, CT
2005 – 2008	Ernest R. Davidson Postdoctoral Fellow Postdoctoral Mentor: Prof. Krishnan Raghavachari Department of Chemistry, Indiana University Bloomington, IN

EDUCATION

2001 – 2005	Ph.D. (Theoretical Chemistry) Wayne State University, Detroit, MI
1997 – 2001	B.S. *cum laude* (Professional Chemistry) Eastern Michigan University, Ypsilanti, MI

SELECTED HONORS & AWARDS

2022	Alumni Achievement Award, Eastern Michigan University
2019	National Science Foundation CAREER Award
2017 – 2018	Hellman Faculty Fellowship, University of California
2005 – 2008	E. R. Davidson Postdoctoral Fellowship, Indiana University
2005	Dan Trivich Memorial Award, Wayne State University

GRANTS AWARDED

Gordon and Betty Moore Foundation. "UC Merced Moore Postdoctoral Fellowship Program", PI: H. P. Hratchian, co-PIs: S. Ghosh, G. Wilson. $1,000,000, 2025–2028. (My participation in this project is through my role as UC Merced's Graduate Dean.)

National Science Foundation. "Venture for Innovation in Self-assembly and integration of Optoelectronic Nanostructures (VISION)", PI: M. Scheibner, co-PIs: H. Cai, S. Ghosh, L. Neshyba, D. Ginger. H. P. Hratchian is Senior Personnel on this award. $1,000,000, 2024–2027 (DMR-2425230).

Mellon Foundation. "Our Interwoven Futures: The Public Humanities Design Studio as a Community-Engaged Hub for Dialogue on Social Justice", PI: C. Lux, co-PIs: I. López-Calvo, R. DeLugan, H. P. Hratchian . $750,000, 2024–2027. (My participation in this project is through my role as UC Merced's Graduate Dean.)

Department of Energy. "Localized electrons, ionization potentials, and non-thermal states in warm dense matter from the thermal projection-based initial maximum overlap method", PI: A. Pribram-Jones, co-PI: H. P. Hratchian. $637,400, 2023–2027 (DE-SC0024476).

Department of Energy. "Exploring photon-initiated electron-neutral interactions in strongly correlated lanthanide complexes", PI: C. C. Jarrold, co-PIs: H. P. Hratchian and L. M. Thompson. $1,356,131 ($443,709 to UCM), 2023–2027 (DE-SC0024282).

National Science Foundation. "CC*Regional Computing: CENVAL-ARC: Central Valley Accessible Research and Computational Hub", PI: S. Chadalapaka, co-PIs: E. J. McTavish, A. Klimaszewski-Patterson, S. Sindi. H. P. Hratchian is Senior Personnel on this award. $1,000,000, 2024–2026 (OAC-2346744).

GRANTS AWARDED, CONTINUED

National Science Foundation. "Planning: PREC Track 2: The UC Merced/MolSSI Chemical Computation and Theory Pathway Program", PI: H. P. Hratchian, co-PIs: C. M. Isborn, A. Pribram-Jones, L. Shi, D. A. Strubbe. $183,978, 2023–2026 (CHE-2335166).

National Science Foundation. "MRI Acquisition of Pinnacles – Raising Research Computing to New Heights in California's Central Valley", PI: H. P. Hratchian, co-PIs: A. M. Martini, S. Sindi. $700,000 (plus $300,000 cost-share funds from UCM), 2020–2023 (ACI-2019144).

National Science Foundation. "CAREER: Development of Efficient Spin Projection Models for Applications to Transition Metal Catalysis", PI: H. P. Hratchian. $625,000, 2019–2024 (CHE-1848580).

Department of Energy. "Improved methods for modeling functional transition metal compounds in complex environments: Ground states, excited states, and spectroscopies", PI: H. P. Hratchian, co-PIs: C. M. Isborn, A. Pribram-Jones, L. Shi, D. A. Strubbe. $1,580,000, 2018–2022 (DE-SC0019053).

National Science Foundation. "Building Capacity: Improving the Undergraduate Chemistry Experience at HSIs by Bridging the GAP (through Green chemistry, Active-learning, and Peer-led experiences)", PI: E. Menke, co-PIs: H. P. Hratchian, C. M. Isborn, J. M. Leslie, B. Stokes. $1,442,497, 2018–2023 (DUE-1832538).

Petroleum Research Fund (ACS). "Computational Studies of Fundamental Organic Transformations Catalyzed by Metal Oxides", PI: H. P. Hratchian. $110,000, 2016–2019 (56806-DNI6).

Hellman Family Foundation. "Exploring the Structure and Chemistry of Metal Oxide Clusters", PI: H. P. Hratchian. $10,000, 2017–2018.

UC Merced Center for Engaged Teaching and Learning. "Theoretical/Computational concept modules for General Chemistry / General Chemistry Honors", PI: H. P. Hratchian, co-PI: E. Menke. $25,000, 2017–2018.

UC Merced Center for Engaged Teaching and Learning. "Writing and Honors General Chemistry - Learning community", PI: P. Gibbons, co-PIs: E. Menke, H. P. Hratchian, A. Zanzucchi. $25,000, 2017–2018.

National Science Foundation. "MRI Acquisition: Multi-Environment Research Computer for Exploration and Discovery (MERCED) Cluster", PI: H. P. Hratchian, co-PIs: J. Q. Sun, S. Sindi. $515,842, 2014–2017 (ACI-1429783).

SELECTED LEADERSHIP, GOVERNANCE, AND PROFESSIONAL SERVICE

University Leadership and Shared Governance: Member, Council of Graduate Schools Committee on Graduate Pathways to Success (2026–Present); Chair, Dean of Natural Sciences Search Committee (2024–2025); Member, Vice Chancellor and Chief Information Officer Search Committee (2022); Chair, Graduate Council (2020–2021); Vice Chair, Graduate Council (2018–2020); UC Merced representative to the UC Systemwide Coordinating Committee on Graduate Affairs (2020–2021); Member, Graduate Council (2016–Present); Member, Campus Budget Working Group (2018–2019); Member, Campus Academic Reorganization Working Group (2017–2018); Member, Campus Instructional Budget Guideline Review Committee (2019–2020); Member, Committee on Research Computing (2015–2022).

Departmental and Academic Program Leadership: Chair and Executive Committee Member, Chemistry & Chemical Biology Graduate Group Education Policy Committee (2015–2020); Faculty Co-Director, Chemistry Center undergraduate learning and tutoring center (2018–2021); Chemical Sciences Faculty Assessment Organizer (2015–2020); Member, Chemistry & Biochemistry Graduate Group Educational Policy Committee (2020–2021).

American Chemical Society Leadership and Governance: ACS COMP Division Councilor (2023–2025) and Alternate Councilor (2026–Present). In this elected role, also serves on the COMP Executive Committee and participates in governance and strategic planning activities for the American Chemical Society.

National Professional Service and Scientific Leadership: Member, ACS COMP National Meeting Program Board; co-organizer of the ACS COMP Standing Symposium on Quantum Mechanics (2018–Present). Organizer of specialized symposia at national ACS meetings, including *Exploring Transition Metal Chemistry & Spectroscopy with Quantum Chemistry* (2019) and the symposium honoring H. B. Schlegel (2013). Served as proposal reviewer and panelist for the National Science Foundation, U.S. Department of Energy, Petroleum Research Fund, and the Center for Nanoscale Materials Proposal Evaluation Board.

Community Engagement and STEM Outreach: Co-developed and presented chemistry outreach programs for local elementary schools in collaboration with UC Merced faculty colleagues. Mentor for the American Chemical Society Project SEED program supporting high school student engagement in STEM research.

PROFESSIONAL AFFILIATIONS

American Chemical Society; American Physical Society; American Association for the Advancement of Science

MENTEES

Postdoctorals: Ali Abou Taka (2025–Present), Hector Corzo (2019–2021; Staff Scientist, Oakridge National Laboratory), Lee Thompson (2014–2017; Associate Prof., Univ. of Louisville)

MENTEES

Graduate Students: Abigail Gyamfi, Brianna Aguilar-Solis, Li Ji, Joseph Kelleher, Amanda Padam, Christian Dywer (M.S., 2026), Cristian Sarabia (Ph.D., 2026), Andrew Bovill (Ph.D., 2026), Abdul Zamani (Ph.D., 2023), Madison Martin (M.S., 2022), Ali Abou Taka (Ph.D., 2021), Samantha Bidwell (Ph.D., 2021), Hassan Harb (Ph.D., 2021), Xianghai Sheng (Ph.D., 2019), Stephen Flaherty (Applied Math, 2016), Lisa Gong (2015–2016)

Undergraduate Students: Michela LoPresti (current), Emma Brass (B.S., 2022), Jonathan Loera (B.S., 2022), Jorge Vazquez (B.S., 2022), Yogev Gluzman (B.S., 2019), Bryce Fairless (B.S., 2019), Anissa Abdullah (B.S., 2019), Preston Griffon, (B.S., 2018), Susana Calderon, (B.S., 2018), Sheyda Partovi (B.S., 2017), Nicole Giddings (B.S., 2015), Stephen Flaherty (B.S., 2015), Euna Chung (B.S., 2015), Ashlee Chan (B.S., 2015), Victor Lee (B.S., 2014), Nicole Degregorio (B.S., 2014)

Summer High School and ACS SEED Students: Shelbie Ernst (2025), Christina Chan (2022), Joanne Guan (2022), Kaneen Muldrow (2019), Leonardo Canchola Colin (2018), Adrian Gomez (2017), Sophia Ortiz (2016), Erika Sanchez (2016), Leonor Alcaraz-Guzman (2015, 2016), James Willis (2014)

PUBLICATIONS

78. S. Nain, M. M. F. de Moraes, H. P. Hratchian, C. C. Jarrold, and L. M. Thompson, "Revisiting the CeO anion photoelectron spectrum: Resolving the assignment and photodetachment mechanism of satellite transitions", *J. Chem. Phys.*, submitted (arXiv: 2609.16657).

77. A. King, A. Rezaei, A. Bovill, R. Semwal, D. Unruh, H. P. Hratchian, A. Gorden, and M. Findlater, "Recyclable homogeneous zinc and cobalt complexes in catalytic hydrosilylation of ketones and amides", *Dalton Trans.*, submitted.

76. L. M. Thompson, H. P. Hratchian, C. C. Jarrold, "Photodetachment spectra of lanthanide oxide clusters: Prevalence and importance of shake-up transitions", *Phys. Chem. Chem. Phys.*, accepted.

75. A. Bovill, A. Abou Taka, H. Harb, and H. P. Hratchian, "Excitation/relaxation analysis of electronic transitions using difference density natural orbitals", *J. Chem. Theory Comput.* **22**, 930–939 (2026).

74. C. D. Huizenga, S. Vaish, C. Dwyer, A. Abou Taka, A. Bovill, L. M. Thompson, H. P. Hratchian, and C. C. Jarrold, "Exploring anomalous photoelectron angular distributions in the photoelectron spectra of $Gd_3O_3^-$: Study of $Gd_3O_2^-$ and $Gd_3O_3^-$ using photoelectron spectroscopy and density functional theory calculations", *J. Phys. Chem. A* **129**, 11607–11623 (2025).

73. A. V. Marenich, E. N. Brothers, H. P. Hratchian, and M. J. Frisch, "Generalized internal coordinates for creative exploration of interatomic geometries", *J. Chem. Theory Comput.* **21**, 10930–10944 (2025).

PUBLICATIONS, CONTINUED

72. V. Rishi, A. Abou Taka, H. P. Hratchian, and L. M. McCaslin, "Quantifying design principles for light-emitting materials with inverted singlet-triplet energy gaps", *J. Phys. Chem. Lett.* **16**, 5213–5220 (2025).

71. A. M. Kinyua, H. P. Hratchian, C. C. Jarrold, and L. M. Thompson, "Photoelectron–remnant interaction effect on remnant wavefunction in low-kinetic energy electron detachment events", *J. Chem. Phys.* **162**, 064304 (2025).

70. A. Y. Zamani and H. P. Hratchian, "Estimating vertical core-excitation energies from Møller-Plesset theory with spin projection", *Mol. Phys.*, e2398142 (2024). Sanibel Conference special issue.

69. S. Vaish, A. O. Gyamfi, C. D. Huizenga, H. P. Hratchian, and C. C. Jarrold, "Electronic structures of small stoichiometric Zn_xO_x clusters", *J. Phys. Chem. A* **128**, 6450–6461 (2024).

68. H. P. Hratchian, A. Karton, and N. J. Mayhall, "Tribute to Krishnan Raghavachari", *J. Phys. Chem. A* **129**, 2523–2525 (2024). *Invited Guest Editorial*.

67. A. Y. Zamani and H. P. Hratchian, "Δ-based composite models for calculating x-ray absorption and emission energies", *J. Chem. Phys.* **159**, 224109 (2023).

66. A. Abou Taka, H. H. Corzo, A. Pribram-Jones, and H. P. Hratchian, "Good Vibrations: Calculating excited state frequencies using ground state self-consistent field models", *J. Chem. Theory Comput.* **8**, 7286–7297 (2022).

65. J. M. Herbert, M. Head-Gordon, H. P. Hratchian, T. Head-Gordon, R. Amaro, A. Aspuru-Guzik, R. Hoffmann, C. Parish, C. M. Payne, and T. Van Voorhis, "Words Matter: On the Debate over Free Speech, Inclusivity, and Academic Excellence", *J. Phys. Chem. Lett.* **13**, 7100–7104 (2022). Guest Commentary

64. A. Y. Zamani and H. P. Hratchian, "Assessing the performance of ΔSCF and the diagonal second-order self-energy approximation for calculating vertical core excitation energies", *J. Chem. Phys.* **157**, 084115 (2022).

63. S. I. Baker, M. Yaghoubi, S. Bidwell, S. L. Pierce, H. P. Hratchian, and R. .D. Baxter, "Enhanced Reactivity for Aromatic Bromination via Halogen-Bonding with Lactic Acid Derivatives", *J. Org. Chem.* **87**, 8492–8502 (2022).

62. A. Abou Taka, S. Lu, D. Gowland, T. J. Zuehlsdorff, H. H. Corzo, A. Pribram-Jones, L. Shi, H. P. Hratchian, and C. M. Isborn, "Comparison of Linear Response Theory, Projected Initial Maximum Overlap Method, and Molecular Dynamics Based Vibronic Spectra: The Case of Methylene Blue", *J. Chem. Theory Comput.* **18**, 3039–3051 (2022).

61. H. Harb and H. P. Hratchian, "A Density Functional Theory Investigation of the Reaction of Water with Ce_2O^-", *Comput. Theor. Chem.* **1209**, 113603 (2022).

PUBLICATIONS, CONTINUED

60. H. H. Corzo, A. Abou Taka, A. Pribram-Jones, and H. P. Hratchian, "Using Projection Operators With Maximum Overlap Methods to Simplify Challenging Self-Consistent Field Optimization", *J. Comp. Chem.* **43**, 382–390 (2022).

59. J. L. Mason, H. Harb, A. Abou Taka, C. D. Huizenga, H. Corzo, H. P. Hratchian, and C. C. Jarrold, "New Photoelectron–Valence Electron Interactions Evident in the Photoelectron Spectrum of Gd_2O^-", *J. Phys. Chem. A* **125**, 9892–9903 (2021).

58. C. D. Huizenga, H. P. Hratchian, and C. C. Jarrold, "Lanthanide Oxides: From Diatomics to High-Spin, Strongly Correlated Homo- and Heterometallic Clusters", *J. Phys. Chem. A* **125**, 6315–6331 (2021). Feature Article.

57. J. D. Galloway, C. Sarabia, J. C. Fettinger, H. P. Hratchian, and R. .D. Baxter, "Versatile New Reagent for Nitrosation under Mild Conditions", *Org. Lett.* **23**, 3253–3258 (2021).

56. H. Harb and H. P. Hratchian, "ΔSCF Dyson Orbitals and Pole Strengths From Natural Ionization Orbitals", *J. Chem. Phys.* **154**, 084104 (2021).

55. J. L. Mason, H. Harb, A. Abou Taka, A. J. McMahon, C. D. Huizenga, H. Corzo, H. P. Hratchian, and C. C. Jarrold, "Photoelectron Spectra of $Gd_2O_2^-$ and Nonmonotonic Photon-Energy-Dependent Variations in Populations of Close-Lying Neutral States", *J. Phys. Chem. A* **125**, 857–866 (2021).

54. A. Abou Taka, M. C. Babin, X. Sheng, J. A. DeVine, D. M. Neumark, and H. P. Hratchian, "Unveiling the Coexistence of Cis and Trans Isomers in the Hydrolysis of ZrO_2: A Coupled DFT and High-Resolution Photoelectron Spectroscopy Study", *J. Chem. Phys.* **153**, 244308 (2020).

53. X. Sheng, L. M. Thompson, and H. P. Hratchian, "Assessing the Calculation of Exchange Coupling Constants and Spin Crossover Gaps Using the Approximate Projection Model To Improve Density Functional Calculations", *J. Chem. Theory Comput.* **16**, 154–163 (2020).

52. J. L. Mason, H. Harb, J. E. Topolski, H. P. Hratchian, and C. C. Jarrold, "Exceptionally Complex Electronic Structures of Lanthanide Oxides and Small Molecules", *Acc. Chem. Res.* **52**, 3265–3273 (2019).

51. H. Harb, L. M. Thompson, and H. P. Hratchian, "On the Linear Geometry of Lanthanide Hydroxides (Ln$-$OH, Ln=La-Lu)", *Phys. Chem. Chem. Phys.* **21**, 21890–21897 (2019).

50. L. C. Reimer, J. M. Leslie, S. .L. Bidwell, C. M. Isborn, D. Lair, E. Menke, B. J. Stokes, and H. P. Hratchian, "Aiming Toward an Effective Hispanic Serving Chemistry Curriculum", In *ACS Symposium Series. Growing Diverse STEM Communities: Methodology, Impact, and Evidence*, edited by L. L. Winfield, G. Thomas, L. M. Watkins, and Z. S. Wilson-Kennedy (American Chemical Society, Washington, D.C.), 29–66 (2019).

PUBLICATIONS, CONTINUED

49. L. M. Thompson, H. P. Hratchian, "On approximate projection models", *Mol. Phys.* **117**, 1421–1429 (2019). (Dieter Cremer Memorial Issue)

48. A. M. Hua, S. L. Bidwell, S. I. Baker, H. P. Hratchian, and R. .D. Baxter, "Experimental and Theoretical Evidence for Nitrogen-Fluorine Halogen Bonding in Silver-Initiated Radical Flourinations", *ACS Catal.* **9**, 3322–3326 (2019).

47. J. L. Mason, H. Harb, C. D. Huizenga, J. C. Ewigleben, J. E. Topolski, H. P. Hratchian, and C. C. Jarrold, "Electronic and Molecular Structures of the CeB_6 Monomer", *J. Phys. Chem. A* **123**, 2040–2048 (2019). (Hanna Reisler Festschrift)

46. X. Cai, A. Tohti, C. Ramirez, H. Harb, J. C. Fettinger, H. P. Hratchian, and B. J. Stokes, "Dispersion-Controlled Regioselective Acid-Catalyzed Intramolecular Hydroindolation of cis-Methindolylstyrenes to Access Tetrahydrobenzo[cd]indoles", *Org. Lett.* **21**, 1574–1577 (2019).

45. J. L. Mason, H. Harb, J. E. Topolski, H. P. Hratchian, and C. C. Jarrold, "A tale of two stabilities: How one boron atom affects a switch in bonding motifs in $CeO_2B_x^-$ (x = 2, 3) complexes", *J. Phys. Chem. A* **122**, 9879–9885 (2018).

44. J. E. Topolski, J. O. Kafader,V. Marrero-Colon, S. S. Iyengar, H. P. Hratchian, and C. C. Jarrold, "Exotic electronic structures of $Sm_xCe_{3-x}O_y$ (x = 0-3; y = 2-4) clusters, and the effect of high neutral density of low-lying states on photodetachment transition intensities", *J. Chem. Phys.* **149**, 054305 (2018).

43. J. A. DeVine, A. Abou Taka, M. C. Babin, M. L. Weichman, H. P. Hratchian, and D. M. Neumark, "High-resolution photoelectron spectroscopy of $TiO_3H_2^-$: Probing the TiO_2^- + H_2O dissociative adduct", *J. Chem. Phys.* **148**, 222810 (2018).

42. L. M. Thompson, C. C. Jarrold, and H. P. Hratchian, "Simulation of low energy photoelectron transitions in $MoVO_4^-$", *J. Chem. Phys.* **146**, 104301 (2017).

41. M. J. Frisch, *et al.*, "GAUSSIAN 16", Gaussian, Inc., Wallingford, CT (2016).

40. A. Petrone, P. Cimino, G. Donati, H. P. Hratchian, M. J. Frisch, and N. Rega, "On the Driving Force of the Excited-State Proton Shuttle in the Green Fluorescent Protein: A Time-Dependent Density Functional Theory (TD-DFT) Study of the Intrinsic Reaction Path", *J. Chem. Theory Comput.* **12**, 4925–4933 (2016).

39. L. M. Thompson, H. Harb, and H. P. Hratchian, "Natural Ionization Orbitals for interpreting electron detachment processes", *J. Chem. Phys.* **144**, 204117 (2016).

PUBLICATIONS, CONTINUED

38. D. Presti, F. Labat, A. Pedone, M. J. Frisch, H. P. Hratchian, I. Ciofini, M. C. Menziani, and C. Adamo, "Modeling emission features of salicylidene aniline molecular crystals: A QM/QM' approach", *J. Comp. Chem.* **37**, 861–870 (2016).

37. S. Cummings, H. P. Hratchian, C. A. Reed, "The strongest acid: Protonation of carbon dioxide", *Angew. Chem. Int. Ed.* **128**, 1404–1408 (2016).

36. L. M. Thompson, H. P. Hratchian, "Modeling the photoelectron spectra of $MoNbO_2^-$ accounting for spin contamination in Density Functional Theory", *J. Phys. Chem. A* **119**, 8744–8751 (2015).

35. L. M. Thompson, H. P. Hratchian, "Second derivatives for approximate spin projection methods", *J. Chem. Phys.* **142**, 054106 (2015).

34. D. Presti, F. Labat, A. Pedone, M. J. Frisch, H. P. Hratchian, M. C. Menziani, I. Ciofini, and C. Adamo, "A computational protocol for modeling thermochromic molecular crystals: salicylidene aniline as a case study", *J. Chem. Theory Comput.* **10**, 5577–5585 (2014).

33. L. M. Thompson, H. P. Hratchian, "Spin projection with double hybrid density functional theory", *J. Chem. Phys.* **141**, 034108 (2014).

32. B. C. Gamoke, U. Das, H. P. Hratchian, and K. Raghavachari, "Divalent pseudoatoms in cluster modeling of Si(100) surfaces", *J. Chem. Phys.* **139**, 164708 (2013).

Publications as Research Scientist at Gaussian, Inc.

31. H. P. Hratchian, "An efficient analytic gradient theory for approximate spin projection methods", *J. Chem. Phys.* **138**, 101101 (2013).

30. H. P. Hratchian and E. Kraka, "Improved predictor-corrector integrators for evaluating reaction path curvature", *J. Chem. Theory Comput.* **9**, 1481-–1488 (2013).

29. H. P. Hratchian and X. S. Li, "Thirty years of geometry optimization in quantum chemistry and beyond: A tribute to Berny Schlegel", *J. Chem. Theory Comput.* **8**, 4853–4855 (2012). *Invited Guest Editorial*.

28. H. P. Hratchian, "Using efficient predictor-corrector reaction path integrators for studies involving projected frequencies", *J. Chem. Theory Comput.* **8**, 5013–5019 (2012).

27. H. P. Hratchian and M. J. Frisch, "Integrating steepest-descent reaction pathways for large molecules", *J. Chem. Phys.* **134**, 204103 (2011).

26. H. P. Hratchian, A. V. Krukau, P. V. Parandekar, K. Raghavachari, M. J. Frisch, and T. Vreven, "QM:QM embedding using electronic densities within and ONIOM framework. Energies and analytic gradients", *J. Chem. Phys.* **135**, 014105 (2011).

PUBLICATIONS, CONTINUED

25. F. Labat, I. Coifini, H. P. Hratchian, K. Raghavachari, M. J. Frisch, and C. Adamo, "Insights into working principles of N_3/TiO_2 dye-sensitized solar cells from first principles modeling", *J. Phys. Chem. C* **115**, 4297–4306 (2011).

24. H. P. Hratchian, M. J. Frisch, and H. B. Schlegel, "Steepest descent reaction path integration using a first-order predictor-corrector method", *J. Chem. Phys.* **133**, 224101 (2010).

23. N. J. Mayhall, K. Raghavachari, and H. P. Hratchian, "ONIOM-based QM:QM electronic embedding method using Löwdin atomic charges: Energies and analytic gradients", *J. Chem. Phys.* **132**, 114107 (2010).

22. J. L. Sonnenberg, H. P. Hratchian, and H. B. Schlegel, "Spin contamination in inorganic chemistry calculations", In *Computational Inorganic and Bioinorganic Chemistry*, edited by E. I. Solomon, R. B. King, and R. A. Scott (Wiley, Chichester, U.K.), 173–186 (2009).

21. F. Labat, I. Ciofini, H. P. Hratchian, M. J. Frisch, K. Raghavachari, and C. Adamo, "First principles modeling of eosin-loaded ZnO films: A step toward the understanding of a dye-sensitized solar cell performances", *J. Am. Chem. Soc.* **131**, 14290–14298 (2009).

20. M. J. Frisch, *et al.*, "GAUSSIAN 09", Gaussian, Inc., Wallingford, CT (2009).

Publications as Student/Postdoc Prior to Independent Career

19. P. V. Parandekar, H. P. Hratchian, and K. Raghavachari, "Applications and assessment of QM:QM electronic embedding using generalized asymmetric Mulliken atomic charges", *J. Chem. Phys.* **129**, 145101 (2008).

18. H. P. Hratchian, P. V. Parandekar, K. Raghavachari, M. J. Frisch, and T. Vreven, "QM:QM electronic embedding using Mulliken atomic charges. Energies and analytic gradients in an ONIOM framework", *J. Chem. Phys.* **128**, 034107 (2008).

17. R. Shakya, S. S. Hindo, L. Wu, M. Allard, M. J. Heeg, H. P. Hratchian, B. R. McGarvey, S. R. P. da Rocha, and C. N. Verani, "Archetypical modeling and amphiphilic behavior of cobalt(II)-containing soft-materials with asymmetric tridentate ligands", *Inorg. Chem.* **46**, 9808–9818 (2007).

16. R. Shakya, S. S. Hindo, L. Wu, S. Ni, M. Allard, M. J. Heeg, S. R. P. da Rocha, G. T. Yee, H. P. Hratchian, and C. N. Verani, "Amphiphilic and magnetic properties of a new class of cluster-bearing $[L_2Cu_4(\mu_4-O)(\mu_2-\text{carboxylato})_4]$ soft materials", *Chem. Eur. J.* **13**, 9948–9956 (2007).

15. H. E. Herbert, M. D. Halls, H. P. Hratchian, and K. Raghavachari, "Hydrogen-bonding interactions in peptide nucleic acid: A comparative study", *J. Phys. Chem. B* **110**, 3336–3343 (2006).

PUBLICATIONS, CONTINUED

14. R. Shakya, C. Imbert, H. P. Hratchian, M. Lanznaster, M. J. Heeg, H. B. Schlegel, B. R. McGarvey, and C. N. Verani, "Structural, spectroscopic, and electrochemical behavior of trans-phenolato cobalt(III) complexes of asymmetric NN'O ligands as archetypes for metallomesogens", *Dalton Trans.* **2006**, 2517–2525 (2006). Selected as the issue's "Hot Article".

13. J. E. Knox, M. D. Halls, H. P. Hratchian, H. B. Schlegel, "Chemical failure modes of AlQ3-based OLEDs: AlQ3 hydrolysis", *Phys. Chem. Chem. Phys.* **8**, 1371–1377 (2006).

12. M. Lanznaster, H. P. Hratchian, M. J. Heeg, L. M. Hryhorczuk, B. R. McGarvey, H. B. Schlegel, and C. N. Verani, "Structural and electronic behavior of unprecedented five-coordinate iron(III) and gallium(III) complexes with a new phenol-rich electroactive ligand", *Inorg. Chem.* **45**, 955–957 (2006).

11. H. P. Hratchian, J. L. Sonnenberg, P. J. Hay, R. L. Martin, B. E. Bursten, and H. B. Schlegel, "Theoretical investigation of uranyldihydroxide: Oxo ligand exchange, water catalysis, and vibrational spectra", *J. Phys. Chem. A* **109**, 8579–8586 (2005).

10. C. Imbert, H. P. Hratchian, M. Lanznaster, M. J. Heeg, L. M. Hryhorczuk, B. R. McGarvey, H. B. Schlegel, and C. N. Verani, "Influence of ligand rigidity and ring substitution on the structural and electronic behavior of trivalent iron and gallium complexes with asymmetric tridentate ligands", *Inorg. Chem.* **44**, 7414–7422 (2005).

9. H. P. Hratchian and H. B. Schlegel, "Finding minima, transition states, and following reaction pathways on ab initio potential energy surfaces", In *Theory and Applications of Computational Chemistry: The First 40 Years*, edited by C. E. Dykstra, G. Frenking, K. S. Kim, and G. E. Scuseria (Elsevier, Amsterdam), 195–249 (2005).

8. H. P. Hratchian and H. B. Schlegel, "Using Hessian updating to increase the efficiency of a Hessian based predictor-corrector reaction path following method", *J. Chem. Theory Comput.* **1**, 61–69 (2005).

7. H. P. Hratchian and M. C. Milletti, "First principles determination of ^{99}Ru chemical shifts using moderately sized basis sets", *J. Mol. Struc.* **724**, 45–52 (2005).

6. H. P. Hratchian, S. K. Chowdury, V. M. Gutiérrez-García, H. B. Schlegel, and J. Montgomery, "Combined experimental and computational investigation of the mechanism of nickel-catalyzed three-component addition processes", *Organometallics* **23**, 4636–4646 (2004). [Addition/Correction: 23, 5652 (2005)]

5. H. P. Hratchian and H. B. Schlegel, "Reaction path following using a Hessian based predictor-corrector algorithm", *J. Chem. Phys.* **120**, 9918–9924 (2004).

PUBLICATIONS, CONTINUED

4. J. E. Knox, H. P. Hratchian, N. Trease, J. Struble, H. B. Schlegel, and H. Holmes, "Using stationary points on potential energy surfaces to model intermolecular interactions and retention in gas chromatography", *Chromatographia* **59**, 329–334 (2004).

3. M. J. Frisch, *et al.*, "GAUSSIAN 03", Gaussian, Inc., Wallingford, CT (2004).

2. H. P. Hratchian and H. B. Schlegel, "Following reaction pathways using a damped classical trajectory algorithm", *J. Phys. Chem. A* **106**, 165–169 (2002).

1. H. P. Hratchian, T. Prendergast, and M. C. Milletti, "Theoretical investigation of substituent effects on the silicon-metal bond for a series of transition metal-substituted base-stabilized silylene complexes", *Polyhedron* **20**, 209–213 (2001).